

UCL DEPARTMENT OF COMPUTER SCIENCE

COMP0080: Graphical Models

Topic-Grouped Past Examination Questions & Exhaustive Solutions Guide

Categorized from 2021–2022, 2023–2024, and Classic Sample Examination Papers

Prepared by **Antigravity AI** for UCL Computer Science

July 2026

Master Course Syllabus & Topic Roadmap

This revision guide categorizes **all past examination questions** for UCL COMP0080 (Graphical Models) by topic, explicitly tagging the source examination year for every question:

1. **2021–22 Final Examination** (COMP0080)
2. **2023–24 Final Examination** (COMP0080A7PF / COMP0080A7UF)
3. **Classic Sample Examination Paper** (COMPGI08 / COMPM056)

Topic Area	Exam Questions & Core Algorithms Covered
Topic 1: Representations	Independence in Joint Tables, Pairwise Independence, Markov Net Separation, State Probabilities, Markov Equivalence, Verma & Pearl Algorithm.
Topic 2: Exact Inference	Factor Graphs, Sum-Product / Belief Propagation, Junction Tree Algorithm (JTA), Moralisation, Triangulation, Running Intersection Property.
Topic 3: Time Series	Hidden Markov Models (HMMs), Forward Filtering (α), Backward Smoothing (β), Likelihood, Prediction, Viterbi Decoding, Exact Sampling.
Topic 4: Learning	Maximum Likelihood Estimation (MLE), Bipartite Markov Nets, Empirical KL Minimization, Randomized Response Surveys, Baum-Welch EM.
Topic 5: Approximate Inference	Matrix KL Divergence, Variational Free Energy ($F(q)$), Mode-Seeking vs Mode-Covering KL, Rejection Sampling, MCMC, Gibbs & MH.
Appendix	Standard Formulas: Chain Rule ($N = 2, 3, \dots, n$), Bayes' Rule, D-Separation, KL Divergence, Jensen's Inequality, Junction Tree RIP, HMM recursions.

TOPIC 1: Representations & Independence in Graphical Models

Question 1.1: Independence in Joint Probability Tables

[Source: 2021–22 Final Exam, Question 1(a)]

Suppose that X_1 and X_2 are two random variables, where each can take one of three states. The joint distribution $P(X_1 = i, X_2 = j)$ is described by matrix P :

$$P = \begin{pmatrix} 0.06 & 0.37 & 0.12 \\ 0.01 & 0.07 & 0.02 \\ 0.03 & 0.26 & 0.06 \end{pmatrix}$$

1. Are X_1 and X_2 independent? Support your answer. [2 Marks]
2. What is the minimal number of entries in the table that you need to change to make them independent while keeping the table a valid probability distribution (non-negative and summing to 1.0)? State the changed entries and the new matrix. [4 Marks]

Solution 1.1

Core Concept Summary: Random variables X_1 and X_2 are statistically independent iff their joint probability matrix P is a rank-1 outer product of marginal vectors $P = rc^T$ (i.e. $P(i, j) = P(X_1 = i)P(X_2 = j)$ for all i, j), which requires all 2×2 subdeterminants of P to be zero.

1. **Exhaustive Verification of Independence:** Calculate row marginal vector $r = [r_1, r_2, r_3]^T$ by summing across columns, and column marginal vector $c = [c_1, c_2, c_3]^T$ by summing across rows:

$$r_1 = 0.06 + 0.37 + 0.12 = 0.55$$

$$r_2 = 0.01 + 0.07 + 0.02 = 0.10$$

$$r_3 = 0.03 + 0.26 + 0.06 = 0.35$$

$$c_1 = 0.06 + 0.01 + 0.03 = 0.10$$

$$c_2 = 0.37 + 0.07 + 0.26 = 0.70$$

$$c_3 = 0.12 + 0.02 + 0.06 = 0.20$$

Notice total sum $\sum_{i,j} P_{ij} = 0.55 + 0.10 + 0.35 = 1.00$. Now test entry $(1, 1)$:

$$r_1 \times c_1 = 0.55 \times 0.10 = 0.055 \neq 0.060 = P_{11}$$

Since $P_{11} \neq r_1 c_1$, the joint distribution does not factorize. Thus, X_1 and X_2 are **not independent**.

2. **Minimal Entries Modification Proof & Derivation:** An independent probability matrix must be rank 1 ($P = rc^T$). Therefore, every 2×2 submatrix of P must have a determinant equal to 0. Let us evaluate the subdeterminants of matrix P :

$$\det \begin{pmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{pmatrix} = \det \begin{pmatrix} 0.06 & 0.37 \\ 0.01 & 0.07 \end{pmatrix} = (0.06)(0.07) - (0.37)(0.01) = 0.0042 - 0.0037 = 0.0005 \neq 0$$

$$\det \begin{pmatrix} P_{22} & P_{23} \\ P_{32} & P_{33} \end{pmatrix} = \det \begin{pmatrix} 0.07 & 0.02 \\ 0.26 & 0.06 \end{pmatrix} = (0.07)(0.06) - (0.02)(0.26) = 0.0042 - 0.0052 = -0.0010 \neq 0$$

Can we make the matrix rank 1 by changing only ****1 single entry****? If we change only entry $P_{22} = x$, setting the first subdeterminant to zero requires $(0.06)(x) - 0.0037 = 0 \implies x = 0.06167$. Setting the second subdeterminant to zero requires $(x)(0.06) - 0.0052 = 0 \implies x = 0.08667$. These requirements conflict, so changing 1 entry is ****impossible****.

Therefore, minimal changes required: **2 entries**. Observe column 1 $[0.06, 0.01, 0.03]^T = 0.10 \times [0.6, 0.1, 0.3]^T$ and column 3 $[0.12, 0.02, 0.06]^T = 0.20 \times [0.6, 0.1, 0.3]^T$. Both columns 1 and 3 share the exact same row ratio vector $[0.6, 0.1, 0.3]^T$. Only Column 2 $[0.37, 0.07, 0.26]^T$ violates this ratio. Setting reference $P_{22} = 0.07$ (so $c_2 = 0.70$), we modify Column 2 to match ratio $[0.6, 0.1, 0.3]^T$: - Change $P_{12} = 0.70 \times 0.6 = \mathbf{0.42}$ (was 0.37). - Keep $P_{22} = 0.70 \times 0.1 = \mathbf{0.07}$ (unchanged). - Change $P_{32} = 0.70 \times 0.3 = \mathbf{0.21}$ (was 0.26).

The new matrix is:

$$P_{\text{new}} = \begin{pmatrix} 0.06 & 0.42 & 0.12 \\ 0.01 & 0.07 & 0.02 \\ 0.03 & 0.21 & 0.06 \end{pmatrix} = \begin{pmatrix} 0.60 \\ 0.10 \\ 0.30 \end{pmatrix} \begin{pmatrix} 0.10 & 0.70 & 0.20 \end{pmatrix}$$

All entries are non-negative, $\sum_{i,j} P_{\text{new},ij} = 1.00$, and the matrix is exactly rank 1. Minimal number of changed entries is **2** (P_{12} and P_{32}).

Question 1.2: Dependencies and Pairwise Independence (XOR)

[Source: 2021–22 Final Exam, Q1(b-c) & 2023–24 Final Exam, Q1(a)]

1. Suppose $A \not\perp B$ and $B \not\perp C$. Does that necessarily imply $A \not\perp C$? Prove or provide a counterexample. [3 Marks]
2. Give an example of a joint distribution in which $A \perp B$, $B \perp C$, $C \perp A$ (pairwise independence), but $A \not\perp B \mid C$. [3 Marks]

Solution 1.2

Core Concept Summary: Statistical dependence is non-transitive ($A \not\perp B \wedge B \not\perp C \not\Rightarrow A \not\perp C$). Pairwise independence between all pairs of variables does not guarantee mutual independence or conditional independence given remaining variables.

1. **Dependency Transitivity Proof & Counterexample:** Statistical dependence is **not transitive**. Let A and C be independent fair coin flips: $A \sim \text{Bernoulli}(0.5)$ and $C \sim \text{Bernoulli}(0.5)$ with $A \perp C$. Define variable B as the composite tuple $B = (A, C)$. - Since B deterministically contains A , knowing B gives full information about $A \Rightarrow A \not\perp B$. - Since B deterministically contains C , knowing B gives full information about $C \Rightarrow B \not\perp C$. - Yet by definition, $A \perp C$. Hence, dependence between A and B , and between B and C , does not imply dependence between A and C .
2. **Pairwise Independence with Conditional Dependence (XOR Gate):** Let $A, B \in \{0, 1\}$ be independent fair binary random variables:

$$P(A = 0) = P(A = 1) = 0.5, \quad P(B = 0) = P(B = 1) = 0.5, \quad P(A, B) = P(A)P(B) = 0.25$$

Define $C = A \oplus B$ (XOR sum modulo 2). $C = 1$ if $A \neq B$ and $C = 0$ if $A = B$.

Check pairwise independence for all pairs:

- Pair (A, B) : $P(A, B) = 0.25 = P(A)P(B) \Rightarrow A \perp B$.
- Pair (B, C) : $P(B = 1, C = 1) = P(B = 1, A = 0) = 0.25 = P(B)P(C) \Rightarrow B \perp C$.
- Pair (A, C) : $P(A = 1, C = 1) = P(A = 1, B = 0) = 0.25 = P(A)P(C) \Rightarrow A \perp C$.

Now evaluate conditional independence $A \perp B \mid C$: Condition on $C = 0$ (which implies $A = B$):

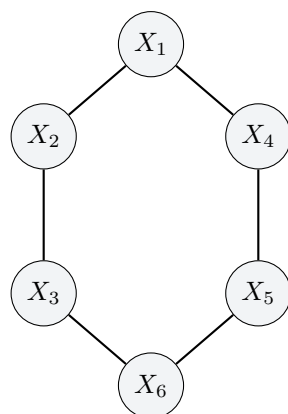
$$P(A = 1, B = 1 \mid C = 0) = \frac{P(A = 1, B = 1, C = 0)}{P(C = 0)} = \frac{0.25}{0.50} = 0.50$$

However, $P(A = 1 \mid C = 0)P(B = 1 \mid C = 0) = 0.50 \times 0.50 = 0.25$. Since $0.50 \neq 0.25$, conditioning on C makes A and B strongly dependent ($A \not\perp B \mid C$).

Question 1.3: Markov Network Marginal Independence

[Source: 2021–22 Final Exam, Q1(d) & 2023–24 Final Exam, Q1(c)]

Consider the undirected Markov Network on variables $X_1, \dots, X_6$:



Write down the set of all conditional and marginal independence statements guaranteed to hold for the marginal distribution $P(X_2, X_4, X_6)$. Justify using graph separation. [6 Marks]

Solution 1.3

Core Concept Summary: Marginalizing out variables from an undirected Markov Network connects all neighboring nodes of each eliminated variable, forming a reduced graph where separation determines valid conditional independence statements.

- Graph Reduction via Elimination of $\{X_1, X_3, X_5\}$:** To construct the marginal distribution $P(X_2, X_4, X_6)$, we integrate out variables X_1, X_3, X_5 :
 - Eliminating X_1 (connected to X_2 and X_4) introduces a direct induced edge (X_2, X_4) .
 - Eliminating X_3 (connected to X_2 and X_6) introduces a direct induced edge (X_2, X_6) .
 - Eliminating X_5 (connected to X_4 and X_6) introduces a direct induced edge (X_4, X_6) .
- Marginal Topology & Separation Analysis:** The resulting marginal graph on $\{X_2, X_4, X_6\}$ forms a fully connected 3-clique (K_3) where every node is directly connected to every other node. Because every pair in $\{X_2, X_4, X_6\}$ is adjacent, no node subset can separate any pair of variables. Therefore, the set of guaranteed independence statements for $P(X_2, X_4, X_6)$ is $\emptyset$ (**the empty set**).

Question 1.4: Exact State Probability in Pairwise Markov Net

[Source: 2023–24 Final Exam, Question 1(b)]

Consider a pairwise Markov Network on 3 binary variables $X_1, X_2, X_3 \in \{0, 1\}$ with joint distribution $P(X) \propto \Phi(X_1, X_2)\Phi(X_2, X_3)\Phi(X_3, X_1)$ where:

$$\Phi(X_i, X_j) = \begin{cases} 2 & \text{if } X_i = X_j \\ 1 & \text{if } X_i \neq X_j \end{cases}$$

Compute the exact probability $P(X_1 = X_2 = X_3)$ of all three variables being in the same state. [5 Marks]

Solution 1.4

Core Concept Summary: In a Markov network, the joint probability of state X is $P(X) = \frac{1}{Z} \prod_c \Phi_c(X_c)$ where partition function $Z = \sum_{X'} \prod_c \Phi_c(X')$ normalizes unnormalized potential weights across all state configurations.

- Exhaustive Table of State Potential Weights:** The state space contains $2^3 = 8$ configurations. Unnormalized weight $W(X) = \Phi(X_1, X_2)\Phi(X_2, X_3)\Phi(X_3, X_1)$:
- Partition Function & Exact Probability Computation:** - Weight sum of equal states: $W_{\text{equal}} = 8 + 8 = 16$. - Weight sum of unequal states: $W_{\text{unequal}} = 6 \times 2 = 12$. - Partition Function: $Z = W_{\text{equal}} + W_{\text{unequal}} = 16 + 12 = 28$.
Exact probability of all three variables being equal:

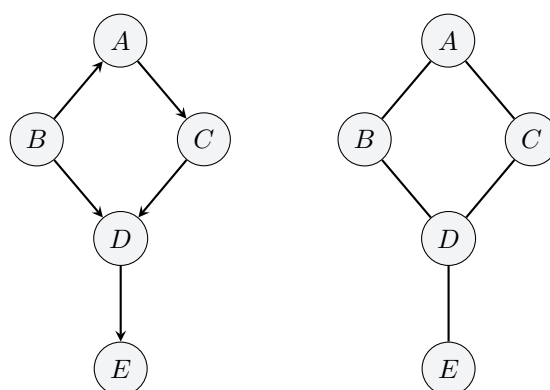
$$P(X_1 = X_2 = X_3) = \frac{W_{\text{equal}}}{Z} = \frac{16}{28} = \frac{4}{7} \approx 0.5714$$

State (X_1, X_2, X_3)	$\Phi(X_1, X_2)$	$\Phi(X_2, X_3)$	$\Phi(X_3, X_1)$	Unnormalized Weight $W(X)$	Classification
(0, 0, 0)	2	2	2	$2 \times 2 \times 2 = 8$	Equal States
(1, 1, 1)	2	2	2	$2 \times 2 \times 2 = 8$	Equal States
(0, 0, 1)	2	1	1	$2 \times 1 \times 1 = 2$	Unequal States
(0, 1, 0)	1	1	2	$1 \times 1 \times 2 = 2$	Unequal States
(1, 0, 0)	1	2	1	$1 \times 2 \times 1 = 2$	Unequal States
(0, 1, 1)	1	2	1	$1 \times 2 \times 1 = 2$	Unequal States
(1, 0, 1)	1	1	2	$1 \times 1 \times 2 = 2$	Unequal States
(1, 1, 0)	2	1	1	$2 \times 1 \times 1 = 2$	Unequal States

Question 1.5: Markov Equivalence & Graph Comparisons

[Source: 2021–22 Final Exam, Q1(e) & 2023–24 Final Exam, Q1(d)]

1. Compare Model 1 (DAG) and Model 2 (Markov Network):



Model 1 (DAG) Model 2 (Markov Network)

Do they encode the same set of independence assumptions? List all differences. [6 Marks]

2. Compare G1 ($A \rightarrow C \leftarrow B$), G2 ($A \leftarrow C \rightarrow B$), G3 ($A \rightarrow C \rightarrow B$), G4 ($A \leftarrow C \leftarrow B$). Which are Markov equivalent? [6 Marks]

Solution 1.5

Core Concept Summary: Two graphs are Markov Equivalent if and only if they encode the exact same set of conditional independence statements. For DAGs, this requires having identical undirected skeletons and identical immoralities (v-structures $X \rightarrow Z \leftarrow Y$ with no edge between X and Y).

1. **Model 1 (DAG) vs Model 2 (Markov Net) Independence Differences:** Both models share the exact same undirected skeleton: $\{(A, B), (A, C), (B, D), (C, D), (D, E)\}$. However, Model 1 contains an immorality at D ($B \rightarrow D \leftarrow C$ with no edge between B and C).

Conditional independence comparison between Model 1 and Model 2:

- **Statement $B \perp\!\!\!\perp C \mid A$:** - **Model 1 (DAG):** Path $B \rightarrow D \leftarrow C$ is blocked at collider D (since D is not conditioned on). Path $B \leftarrow A \rightarrow C$ is blocked by conditioning on A . Thus $B \perp\!\!\!\perp C \mid A$ is **TRUE**. - **Model 2 (Markov Net):** Conditioning on A does not block path $B - D - C$. Graph separation fails. Thus $B \perp\!\!\!\perp C \mid A$ is **FALSE**.
- **Statement $B \perp\!\!\!\perp C \mid \{A, D\}$:** - **Model 1 (DAG):** Conditioning on collider D opens the v-structure path $B \rightarrow D \leftarrow C$. Thus $B \perp\!\!\!\perp C \mid \{A, D\}$ is **FALSE**. - **Model 2 (Markov Net):** Conditioning on $\{A, D\}$ removes all paths between B and C . Graph separation holds. Thus $B \perp\!\!\!\perp C \mid \{A, D\}$ is **TRUE**.

Since their conditional independence sets differ, Model 1 and Model 2 are **not Markov Equivalent**.

2. **3-Node DAG Markov Equivalence Evaluation:** All four graphs share the same undirected skeleton $A - C - B$.
 - **G1 ($A \rightarrow C \leftarrow B$):** Has an immorality at C . $A \perp\!\!\!\perp B$ holds, but $A \not\perp\!\!\!\perp B \mid C$.

- **G2** ($A \leftarrow C \rightarrow B$), **G3** ($A \rightarrow C \rightarrow B$), **G4** ($A \leftarrow C \leftarrow B$): None of these graphs contain an immorality. In all three graphs, $A \perp\!\!\!\perp B \mid C$ holds and $A \not\perp\!\!\!\perp B$.

Therefore, **G2**, **G3**, and **G4** are **Markov Equivalent** to each other, while G1 forms its own distinct equivalence class.

Question 1.6: Fundamental Definitions & Universal Factorizations

[Source: Classic Sample Exam, Question 1(a-b)]

1. Explain what dependency between discrete random variables X_i and X_j means mathematically. [2 Marks]
2. Define a Belief Network. Why must its graphical structure be a Directed Acyclic Graph (DAG)? [2 Marks]
3. Prove why any joint distribution $p(x_1, \dots, x_N)$ can be written as a Belief Network. [2 Marks]
4. Explain why the lack of an edge in a DAG implies an independence statement, whereas the presence of an edge does not guarantee dependence. [3 Marks]

Solution 1.6

Core Concept Summary: Belief Networks parameterize joint distributions via recursive conditional probability tables (CPTs) over DAG topologies. DAG acyclicity ensures valid probabilistic conditioning without circular dependence.

1. **Mathematical Dependency Definition:** Variables X_i and X_j are dependent if their joint distribution cannot be factorized into the product of their marginals:

$$p(x_i, x_j) \neq p(x_i)p(x_j) \iff p(x_i \mid x_j) \neq p(x_i)$$

2. **Belief Network Definition & DAG Acyclicity Requirement:** A Belief Network (Bayesian Network) is a directed graphical model that factorizes a joint distribution as:

$$p(x_1, \dots, x_N) = \prod_{i=1}^N p(x_i \mid \text{pa}(x_i))$$

The graph must be a DAG to guarantee that the factorized product represents a valid, normalized probability distribution. Directed cycles (e.g. $A \rightarrow B \rightarrow A$) create circular conditional definitions $p(a \mid b)p(b \mid a)$, which generally do not integrate or sum to 1.

3. **Universal Factorization Proof via Chain Rule:** By the chain rule of probability, any joint distribution over N variables can be expanded without loss of generality as:

$$p(x_1, x_2, \dots, x_N) = p(x_1)p(x_2 \mid x_1)p(x_3 \mid x_1, x_2) \dots p(x_N \mid x_1, \dots, x_{N-1})$$

This expansion directly matches a fully connected DAG where node x_i has parents $\text{pa}(x_i) = \{x_1, \dots, x_{i-1}\}$. Hence, any joint distribution can be represented as a Belief Network.

4. **Lack of Edge vs Presence of Edge:** - **Lack of Edge:** Omitting an edge $X_i \rightarrow X_j$ asserts a structural independence constraint ($X_j \perp\!\!\!\perp X_i \mid \text{pa}(X_j) \setminus \{X_i\}$) that must hold for *all* parameter choices of CPTs. - **Presence of Edge:** An edge $X_i \rightarrow X_j$ allows conditional dependence, but does not guarantee it. For specific numerical choices of CPT parameters (e.g. uniform CPT $p(x_j \mid x_i) = \text{const}$), X_j may remain independent of X_i .

Question 1.7: DAG to Undirected Conversion Counterexample

[Source: Classic Sample Exam, Question 1(b.iii)]

Let DAG A have adjacency matrix A_{ij} . Form undirected adjacency matrix $B_{ij} = A_{ij} \vee A_{ji}$. Can every Belief Network with DAG A be represented by a Markov Network with adjacency B ? Explain or give counterexample. [3 Marks]

Solution 1.7

Core Concept Summary: Converting a DAG to a Markov Network by simply stripping arrowheads (without moralizing unmarried parents) fails for collider structures $A \rightarrow B \leftarrow C$, because the collider induces marginal independence ($A \perp\!\!\!\perp C$) which is violated in the un-moralized undirected graph.

No. Consider the collider DAG $A \rightarrow B \leftarrow C$. - In the DAG, A and C are marginally independent ($A \perp\!\!\!\perp C$). - Stripping arrowheads yields undirected skeleton B with edges (A, B) and (B, C) , but no edge (A, C) . - In undirected graph B , path $A - B - C$ connects A and C , implying $A \not\perp\!\!\!\perp C$. Thus undirected graph B cannot represent the DAG without adding moral edge (A, C) .

Question 1.8: Verma & Pearl Markov Equivalence Algorithm

[Source: Classic Sample Exam, Question 1(c)]

1. Define Markov Equivalence between two graphs. [2 Marks]
2. State an algorithm to determine if two DAGs are Markov equivalent. [2 Marks]
3. Check equivalence for Pair 1: $(a \rightarrow b \rightarrow d, a \rightarrow c \rightarrow d)$ vs $(a \leftarrow b \rightarrow d, a \rightarrow c \rightarrow d)$. [3 Marks]
4. Check equivalence for Pair 2: $(a \rightarrow b \rightarrow d, a \rightarrow c \rightarrow d)$ vs $(a \rightarrow b \rightarrow d, a \rightarrow c \leftarrow d)$. [3 Marks]

Solution 1.8

Core Concept Summary: Verma & Pearl Theorem: Two DAGs are Markov Equivalent if and only if they have the exact same undirected skeleton and the exact same set of immoralities.

1. **Markov Equivalence Definition:** Two graphs G_1 and G_2 are Markov Equivalent if they encode identical sets of conditional independence statements.
2. **Verma & Pearl Algorithm:**
 - (i) Construct the undirected skeleton for both DAGs (strip arrowheads). If skeletons differ, return **Not Equivalent**.
 - (ii) Identify all immoralities (unmarried colliders $X \rightarrow Z \leftarrow Y$ with no edge between X and Y). If the sets of immoralities differ, return **Not Equivalent**. Otherwise return **Markov Equivalent**.
3. **Pair 1 Evaluation:** Both DAGs have skeleton $\{(a, b), (a, c), (b, d), (c, d)\}$ and 0 immoralities. **Markov Equivalent**.
4. **Pair 2 Evaluation:** DAG 1 has no immoralities. DAG 2 has immorality at d ($c \rightarrow d \leftarrow b$). **Not Markov Equivalent**.

TOPIC 2: Exact Inference & Factor Graphs

Question 2.1: The Junction Tree Algorithm

[Source: 2021–22 Final Exam, Q2.2 & 2023–24 Final Exam, Q2(b)]

Consider the joint distribution that factorizes as:

$$p(a, b, c, d, e, f) = p(a)p(b | a)p(c | b) \\ \times p(d | c)p(e | d)p(f | a, e)$$

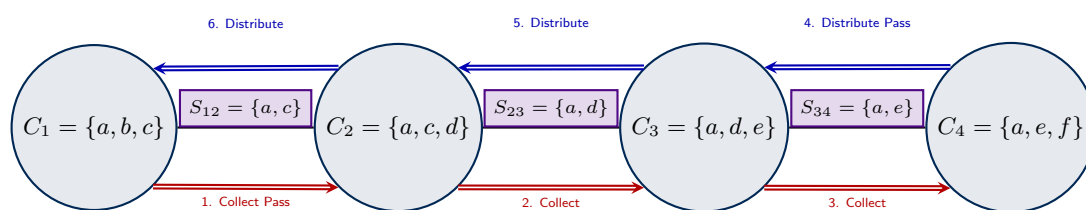
1. Draw the DAG representing this distribution. [3 Marks]
2. Draw the moralised version of this graph. Explain parent marrying. [3 Marks]
3. Draw the triangulated version of the graph with the smallest possible maximum clique size. [3 Marks]
4. Construct a Junction Tree and verify the Running Intersection Property (RIP). [3 Marks]
5. Describe potential initialization, absorption, and message schedule. [3 Marks]

Solution 2.1

Core Concept Summary: The Junction Tree Algorithm converts arbitrary DAGs/Markov networks into a tree of maximal cliques where message absorption guarantees exact marginal probabilities for all variable subsets in $O(N)$ time.

1. **DAG Construction:** Directed edges: $a \rightarrow b \rightarrow c \rightarrow d \rightarrow e$, $a \rightarrow f$, $e \rightarrow f$.
2. **Moralisation & Parent Marrying:** f has parents a and e . Marry parents with moral edge (a, e) and strip arrowheads. Resulting undirected graph has edges $\{(a, b), (b, c), (c, d), (d, e), (a, e), (a, f), (e, f)\}$.
3. **Triangulation:** Undirected 5-cycle $a - b - c - d - e - a$ is triangulated with chords (a, c) and (a, d) . Maximal cliques of size 3: $C_1 = \{a, b, c\}$, $C_2 = \{a, c, d\}$, $C_3 = \{a, d, e\}$, $C_4 = \{a, e, f\}$.
4. **Junction Tree & RIP Verification:** Tree topology: $C_1 \xrightarrow{S_{12}=\{a,c\}} C_2 \xrightarrow{S_{23}=\{a,d\}} C_3 \xrightarrow{S_{34}=\{a,e\}} C_4$. RIP check: Variable a appears in C_1, C_2, C_3, C_4 , forming contiguous sub-tree $C_1 - C_2 - C_3 - C_4$. RIP holds!
5. **Potential Initialization & Message Passing Schedule:** - Initialize clique potentials: $\psi(C_1) = p(a)p(b|a)p(c|b)$, $\psi(C_2) = p(d|c)$, $\psi(C_3) = p(e|d)$, $\psi(C_4) = p(f|a, e)$. Separators $\phi(S) = 1$.

Diagrammatic Junction Tree Message Absorption Schedule



Absorption update rule: $\phi(S_{ij})^* = \sum_{C_i \setminus S_{ij}} \psi(C_i)$, then $\psi(C_j)^* = \psi(C_j) \frac{\phi(S_{ij})^*}{\phi(S_{ij})^{\text{old}}}$.

Question 2.2: Sum-Product Algorithm on Factor Graphs

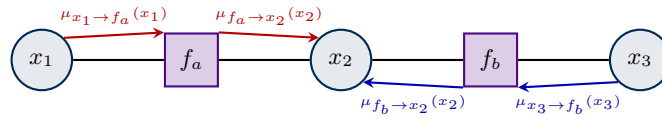
[Source: Classic Sample Exam, Question 2(a)]

Describe the sum-product algorithm for computing marginals in a factor graph. Explain which factor graph topologies the algorithm is applicable to. [4 Marks]

Solution 2.2

Core Concept Summary: The Sum-Product algorithm computes exact marginals on tree-structured factor graphs via two-way message passing between variable nodes x_i and factor nodes f_a .

Diagrammatic Factor Graph Message Passing



Sum-Product message update equations:

$$\mu_{f \rightarrow i}(x_i) = \sum_{\chi_f \setminus \{i\}} f(x_i, \chi_f) \prod_{j \in \text{ne}(f) \setminus \{i\}} \mu_{j \rightarrow f}(x_j)$$

$$\mu_{i \rightarrow f}(x_i) = \prod_{g \in \text{ne}(i) \setminus \{f\}} \mu_{g \rightarrow i}(x_i)$$

Applicable for exact marginals on single-tree factor graphs (no loops).

TOPIC 3: Time Series & Hidden Markov Models (HMMs)

Question 3.1: HMM Posterior Graph & Exact Sampling

[Source: 2021–22 Final Exam, Q2(b.i-ii) & Classic Sample Exam, Q2(b.vii)]

1. When sequence $v_{1:T}$ is observed, it induces posterior distribution $p_v(h_{1:T}) = p(h_{1:T} | v_{1:T})$. What is the graphical model of this posterior? Is $h_1 \perp\!\!\!\perp h_T$? [3 Marks]
2. Can we efficiently sample from $p_v(h_{1:T})$ exactly? Describe how. [3 Marks]

Solution 3.1

Core Concept Summary: Conditioning an HMM on observations $v_{1:T}$ induces an undirected linear chain posterior over hidden states $h_{1:T}$ where h_1 and h_T remain dependent.

1. **Posterior Topology & Dependency:** Linear chain graph $h_1 - h_2 - \dots - h_T$. h_1 and h_T are dependent.
2. **Forward-Filtering Backward-Sampling Algorithm:**
 - (i) Compute forward messages $\alpha_t(h_t) = p(h_t, v_{1:t})$ for $t = 1 \dots T$.
 - (ii) Sample $h_T^* \sim p(h_T | v_{1:T}) \propto \alpha_T(h_T)$.
 - (iii) Step backward for $t = T - 1 \dots 1$: sample $h_t^* \sim p(h_t | h_{t+1}^*, v_{1:t}) \propto p(h_{t+1}^* | h_t) \alpha_t(h_t)$.

Question 3.2: Deterministic State Decoupling & Step-by-Step Filtering

[Source: 2021–22 Final Exam, Question 2.3]

Given $v_t \in \{0, 1, 2\}$, $h_t \in \{0, 1\}$, $p(h_1 = 1) = 0.5$, transition matrix $T = \begin{pmatrix} 0.5 & 0.8 \\ 0.5 & 0.2 \end{pmatrix}$, emission matrix

$B = \begin{pmatrix} 0.3 & 0.5 \\ 0.6 & 0 \\ 0.1 & 0.5 \end{pmatrix}$, and sequence $v_{1:10} = [0, 1, 0, 2, 0, 2, 1, 0, 2, 0]$:

1. Is $h_1 \perp\!\!\!\perp h_{10} | v_{1:10}$ for this specific sequence? Why? [2 Marks]
2. Compute exact filtering distributions $p(h_8 | v_{1:8})$ and $p(h_9 | v_{1:9})$. [3 Marks]

Solution 3.2

Core Concept Summary: Zero-emission probabilities ($B_{v,h} = 0$) pin hidden states deterministically, acting as hard observed split nodes that d-separate past and future hidden states in the chain.

1. **Deterministic State Decoupling Analysis:** Emission row 1 for state $h = 1$ is $B_{1,1} = p(v = 1 | h = 1) = 0$. Since $v_2 = 1$ and $v_7 = 1$, states are pinned deterministically: $h_2 = 0$ and $h_7 = 0$ with probability 1. Conditioning on $h_7 = 0$ d-separates h_1 from h_{10} . They are **independent**.
2. **Filtering Calculations for $t = 8, 9$:**
 - (a) Base at $t = 7$: $p(h_7 | v_{1:7}) = \begin{pmatrix} 1.0 \\ 0.0 \end{pmatrix}$.
 - (b) Predict $t = 8$: $p(h_8 | v_{1:7}) = Tp(h_7 | v_{1:7}) = \begin{pmatrix} 0.5 & 0.8 \\ 0.5 & 0.2 \end{pmatrix} \begin{pmatrix} 1.0 \\ 0.0 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$.
 - (c) Observe $v_8 = 0$: Unnormalized $\begin{pmatrix} 0.3 \times 0.5 \\ 0.5 \times 0.5 \end{pmatrix} = \begin{pmatrix} 0.15 \\ 0.25 \end{pmatrix} \Rightarrow p(h_8 | v_{1:8}) = \begin{pmatrix} 0.375 \\ 0.625 \end{pmatrix} = \begin{pmatrix} 3/8 \\ 5/8 \end{pmatrix}$.
 - (d) Predict $t = 9$: $p(h_9 | v_{1:8}) = Tp(h_8 | v_{1:8}) = \begin{pmatrix} 0.5 & 0.8 \\ 0.5 & 0.2 \end{pmatrix} \begin{pmatrix} 0.375 \\ 0.625 \end{pmatrix} = \begin{pmatrix} 0.6875 \\ 0.3125 \end{pmatrix}$.
 - (e) Observe $v_9 = 2$: Unnormalized $\begin{pmatrix} 0.1 \times 0.6875 \\ 0.5 \times 0.3125 \end{pmatrix} = \begin{pmatrix} 0.06875 \\ 0.15625 \end{pmatrix} \Rightarrow p(h_9 | v_{1:9}) = \begin{pmatrix} 11/36 \\ 25/36 \end{pmatrix} \approx \begin{pmatrix} 0.3056 \\ 0.6944 \end{pmatrix}$.

Question 3.3: HMM Filtering, Smoothing & Viterbi Recursions

[Source: Classic Sample Exam, Question 2(b.i-vi)]

1. Derive forward filtering $\alpha_t(h_t) = p(h_t, v_{1:t})$. [3 Marks]
2. Derive backward smoothing $\beta_t(h_t) = p(v_{t+1:T} | h_t)$. [3 Marks]
3. Compute sequence likelihood $p(v_{1:T})$. [5 Marks]
4. Derive prediction $p(v_{T+1} | v_{1:T})$. [5 Marks]
5. Derive Viterbi algorithm for $\operatorname{argmax}_{h_{1:T}} p(h_{1:T} | v_{1:T})$. [3 Marks]

Solution 3.3

Core Concept Summary: HMM dynamic programming passes forward messages α_t and backward messages β_t to compute marginal posteriors, likelihoods, and Viterbi MAP state paths.

1. **Forward Filtering** $\alpha_t(h_t)$: $\alpha_t(h_t) = p(v_t | h_t) \sum_{h_{t-1}} p(h_t | h_{t-1}) \alpha_{t-1}(h_{t-1})$.
2. **Backward Smoothing** $\beta_t(h_t)$: $\beta_t(h_t) = \sum_{h_{t+1}} p(v_{t+1} | h_{t+1}) p(h_{t+1} | h_t) \beta_{t+1}(h_{t+1})$.
3. **Sequence Likelihood**: $p(v_{1:T}) = \sum_{h_T} \alpha_T(h_T)$.
4. **One-step Ahead Prediction**: $p(v_{T+1} | v_{1:T}) = \sum_{h_{T+1}} p(v_{T+1} | h_{T+1}) \sum_{h_T} p(h_{T+1} | h_T) p(h_T | v_{1:T})$.
5. **Viterbi Max-Product Recursion**: $V_t(h_t) = p(v_t | h_t) \max_{h_{t-1}} [p(h_t | h_{t-1}) V_{t-1}(h_{t-1})]$.

TOPIC 4: Parameter Learning & EM Algorithm

Question 4.1: Bipartite Disease-Symptom Markov Network

[Source: 2021–22 Final Exam, Question 2.1]

Given $p(s, d) = \frac{1}{Z} \exp(s^T W d + a^T s + b^T d)$:

1. Draw the Markov Network. Describe structure. [3 Marks]
2. Derive $p(s \mid d)$. Is it factorized? [4 Marks]
3. Derive log-likelihood L for N records (s^n, d^n) . [3 Marks]
4. Derive gradients $\frac{\partial L}{\partial W_{ij}}, \frac{\partial L}{\partial a_i}, \frac{\partial L}{\partial b_j}$. [5 Marks]
5. Efficiency in general case? [3 Marks]

Solution 4.1

Core Concept Summary: Conditioning a bipartite Markov network on one layer (d) breaks all horizontal ties, causing the conditional distribution over symptoms $p(s \mid d)$ to factorize completely into independent Bernoulli distributions.

1. **Graph Structure:** Bipartite graph between s_i and d_j .
2. **Conditional Distribution** $p(s \mid d)$: $p(s \mid d) = \prod_{i=1}^S \sigma(W_i \cdot d + a_i)^{s_i} (1 - \sigma(W_i \cdot d + a_i))^{1-s_i}$. Fully factorized.
3. **Log-Likelihood Formulation:** $L = \sum_{n=1}^N [(s^n)^T W d^n + a^T s^n + b^T d^n] - N \log Z(W, a, b)$.
4. **Gradient Derivations:** $\frac{\partial L}{\partial W_{ij}} = \sum_{n=1}^N (s_i^n d_j^n - \langle s_i d_j \rangle_{p(s, d)})$.
5. **Computational Efficiency:** Infeasible in general case (2^{S+D} sum for partition function Z).

Question 4.2: Empirical KL Minimization & Edge Removal

[Source: 2023–24 Final Exam, Question 2.2.1]

Belief Net $A \rightarrow B, A \rightarrow C$. Learning parameters by Maximum Likelihood minimizes $KL(Q \parallel P)$ where Q is empirical distribution and P is model distribution.

1. Optimal setting of distribution P and value of $KL(Q \parallel P)$? [5 Marks]
2. Remove edge $A \rightarrow B$: impact on optimal KL divergence? [5 Marks]
3. Dataset conditions for equal KL divergence? [5 Marks]

Solution 4.2

Core Concept Summary: Maximum likelihood parameter estimation in Belief Networks is equivalent to minimizing forward KL divergence $KL(Q \parallel P)$ between empirical data Q and model distribution P .

1. **Optimal Setting:** $P^*(A, B, C) = Q(A, B, C) \implies KL(Q \parallel P^*) = 0$.
2. **Edge Removal Impact:** Constrains P to factorizations $P(A)P(B)P(C|A)$. Minimum KL divergence **cannot decrease** ($KL(Q \parallel P^*) \geq 0$).
3. **Identical KL Dataset Condition:** Holds iff A and B are **marginally independent** under empirical data $Q(A, B) = Q(A)Q(B)$.

Question 4.3: Randomized Response Survey Procedure & EM

[Source: 2021–22 Final Exam, Q2.4 & 2023–24 Final Exam, Q2(c)]

1. Graphical model & CPTs. [2 Marks]
2. Truthfulness posterior $P(C_1 = 1 \mid Y = 1)$ for $p_1 = 0.5$. [2 Marks]
3. MLE for $p_1 = 0.5$. [2 Marks]
4. EM updates for unknown (p_1, p_2) . [2 Marks]

5. Uniqueness for $D = (\text{yes}, \text{yes}, \text{yes}, \text{no})$? [2 Marks]

Solution 4.3

Core Concept Summary: Randomized response adds hidden coin-flip noise C_1 to protect participant privacy, requiring the Expectation-Maximization (EM) algorithm to estimate unobserved parameters.

1. **Graphical Structure:** $H \rightarrow Y \leftarrow C_1$. $P(Y = 1 | H, C_1) = C_1 H + (1 - C_1) \cdot 0.5$.
2. **Truthfulness Posterior:** $P(C_1 = 1 | Y = 1) = \frac{2p_2}{2p_2 + 1}$.
3. **Closed-Form MLE:** $\hat{p}_2 = \max(0, \min(1, 2\frac{n_y}{n} - 0.5))$.
4. **EM Algorithm Steps:**
 - **E-step:** $\langle C_{1,i} \rangle$ and $\langle H_i \rangle$ posteriors.
 - **M-step:** $p_1^{\text{new}} = \frac{1}{n} \sum_i \langle C_{1,i} \rangle$, $p_2^{\text{new}} = \frac{1}{n} \sum_i \langle H_i \rangle$.
5. **Non-Uniqueness Analysis:** $\hat{\theta} = 0.75 \implies p_1(p_2 - 0.5) = 0.25$. 1D solution manifold. **Not unique.**

Question 4.4: Baum-Welch EM Derivation & Monotonicity Proof

[Source: Classic Sample Exam, Question 2(c)]

1. Derive Baum-Welch E and M steps for transition A and emission B . [6 Marks]
2. Prove log-likelihood is monotonically non-decreasing using free energy bound. [3 Marks]

Solution 4.4

Core Concept Summary: Baum-Welch EM maximizes HMM marginal log-likelihood $\log p(V | \theta)$ by iteratively calculating expected transition/emission counts in the E-step and updating A, B matrices in the M-step.

1. **Baum-Welch Updates:** - **E-step:** $\gamma_t^n(i) = \frac{\alpha_t^n(i)\beta_t^n(i)}{p(v_{1:T}^n)}$ and $\xi_t^n(j, i) = \frac{\alpha_{t-1}^n(j)A_{ij}B_{v_t^n, i}\beta_t^n(i)}{p(v_{1:T}^n)}$. - **M-step:** $\hat{A}_{ij} = \frac{\sum_n \sum_t \xi_t^n(j, i)}{\sum_n \sum_t \gamma_{t-1}^n(j)}$ and $\hat{B}_{ki} = \frac{\sum_n \sum_{t: v_t=k} \gamma_t^n(i)}{\sum_n \sum_t \gamma_t^n(i)}$.
2. **Monotonicity Proof:** Log-likelihood decomposition $\log p(V | \theta) = \mathcal{B}(q, \theta) + KL(q \parallel p)$. E-step sets $KL = 0$, M-step maximizes $\mathcal{B}(q, \theta)$, ensuring $\log p(V | \theta^{\text{new}}) \geq \log p(V | \theta^{\text{old}})$.

TOPIC 5: Approximate Inference & Sampling

Question 5.1: Matrix KL Divergence Computation

[Source: 2021–22 Final Exam, Question 3.1]

Given $P(X) = (0.25, 0.25, 0.25, 0.25)^T$, $Q(X) = (0.5, 0.25, 0.125, 0.125)^T$, and matrices P, Q :

$$P = \begin{pmatrix} 0.25 & 0.5 & 0 & 0.2 \\ 0.25 & 0 & 0.5 & 0.3 \\ 0.25 & 0 & 0 & 0.1 \\ 0.25 & 0.5 & 0.5 & 0.4 \end{pmatrix}, \quad Q = \begin{pmatrix} 0.1 & 0 & 0 & 1 \\ 0 & 0 & 0.5 & 0 \\ 0.9 & 0.5 & 0 & 0 \\ 0 & 0.5 & 0.5 & 0 \end{pmatrix}$$

Compute marginals $P(Y), Q(Y)$, and divergences $KL(P(Y) \parallel Q(Y))$ and $KL(Q(Y) \parallel P(Y))$. [5 Marks]

Solution 5.1

Core Concept Summary: KL divergence $KL(P \parallel Q) = \sum P(y) \log \frac{P(y)}{Q(y)}$ is an asymmetric information-theoretic distance measure ($KL(P \parallel Q) \neq KL(Q \parallel P)$).

- Marginals Computation:** $P(Y) = [0.2375, 0.2625, 0.0875, 0.4125]^T$, $Q(Y) = [0.1750, 0.0625, 0.5750, 0.1875]^T$.
- Forward KL $KL(P \parallel Q)$:**

$$\begin{aligned} KL(P \parallel Q) &= 0.2375 \ln \left(\frac{0.2375}{0.1750} \right) + 0.2625 \ln \left(\frac{0.2625}{0.0625} \right) + 0.0875 \ln \left(\frac{0.0875}{0.5750} \right) + 0.4125 \ln \left(\frac{0.4125}{0.1875} \right) \\ &= \mathbf{0.6097 \text{ nats}} \quad (0.8797 \text{ bits}) \end{aligned}$$

- Reverse KL $KL(Q \parallel P)$:**

$$\begin{aligned} KL(Q \parallel P) &= 0.1750 \ln \left(\frac{0.1750}{0.2375} \right) + 0.0625 \ln \left(\frac{0.0625}{0.2625} \right) + 0.5750 \ln \left(\frac{0.5750}{0.0875} \right) + 0.1875 \ln \left(\frac{0.1875}{0.4125} \right) \\ &= \mathbf{0.7916 \text{ nats}} \quad (1.1420 \text{ bits}) \end{aligned}$$

Question 5.2: Mean-Field Variational Free Energy Derivation

[Source: 2021–22 Final Exam, Q3.2 & 2023–24 Final Exam, Q3(a)]

For 3-node pairwise Markov network $P(X) \propto \prod_{i \sim j} \Phi(X_i, X_j)$ where $\Phi(X_i, X_j) = 2$ if $X_i = X_j$ and 1 if $X_i \neq X_j$, derive explicit free energy $F(q)$ under fully factorized $q(X) = \prod_{i=1}^3 q_i(X_i)$. [5 Marks]

Solution 5.2

Core Concept Summary: Mean-field variational inference approximates complex joint distributions with fully factorized distributions $q(X) = \prod q_i(X_i)$ by minimizing variational free energy $F(q)$.

$$F(q) = \sum_{i=1}^3 [\mu_i \log \mu_i + (1 - \mu_i) \log(1 - \mu_i)] - \log 2 \sum_{(i,j)} [\mu_i \mu_j + (1 - \mu_i)(1 - \mu_j)]$$

Question 5.3: Mode-Seeking vs Mode-Covering KL Divergence

[Source: Classic Sample Exam, Question 3(a)]

Explain the operational difference between minimizing $KL(P \parallel Q)$ vs $KL(Q \parallel P)$ when approximating bimodal target P with unimodal proposal Q . [5 Marks]

Solution 5.3

Core Concept Summary: Forward KL $KL(P \parallel Q)$ is zero-avoiding (mode-covering), forcing Q to spread across all modes. Reverse KL $KL(Q \parallel P)$ is zero-forcing (mode-seeking), locking Q onto a single mode.

- Mode-Seeking ($KL(Q \parallel P)$):** Zero-forcing. Q contracts to cover a single peak of P .
- Mode-Covering ($KL(P \parallel Q)$):** Zero-avoiding. Q spreads wide to cover all modes of P .

Question 5.4: Rejection Sampling on Lattices

[Source: 2021–22 Final Exam, Q3(b) & 2023–24 Final Exam, Q3(b)]

Compute exact average acceptance rates $P(\text{accept}) = Z/2^{MN}$ for:

1. $M = 2, N = 1$: $Z = 2 + 2e^{-1} \approx 2.7358 \implies P(\text{accept}) = \mathbf{0.6839}$ (68.39%).
2. $M = 2, N = 2$: $Z = 2 + 12e^{-2} + 2e^{-4} \approx 3.6606 \implies P(\text{accept}) = \mathbf{0.2288}$ (22.88%).

Question 5.5: MCMC, Gibbs & Metropolis-Hastings

[Source: 2021–22 Final Exam, Q3(c), 2023–24 Final Exam, Q3(c) & Classic Sample Exam, Q3(b)]

Detailed balance ($P(x)T(x'|x) = P(x')T(x|x')$) is sufficient for stationary distribution P , but not nec-

essary. Counterexample: Lazy cyclic 3-state chain $T = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \end{pmatrix}$ with $P = [1/3, 1/3, 1/3]^T$.

$PT = P$ holds, but $P(1)T(2|1) = 1/6 \neq 0 = P(2)T(1|2)$. Detailed balance is violated.

APPENDIX: Standard Formulas & Foundational Identities

1. Chain Rule of Probability

The Chain Rule decomposes any joint probability distribution over multiple variables into a product of conditional probabilities.

2 Variables ($N = 2$):

$$p(x_1, x_2) = p(x_1)p(x_2 | x_1) = p(x_2)p(x_1 | x_2)$$

3 Variables ($N = 3$):

$$p(x_1, x_2, x_3) = p(x_1)p(x_2 | x_1)p(x_3 | x_1, x_2)$$

General N Variables ($N \geq 2$):

$$p(x_1, x_2, \dots, x_N) = p(x_1) \prod_{i=2}^N p(x_i | x_1, x_2, \dots, x_{i-1})$$

2. Bayes' Rule & Law of Total Probability

Bayes' Rule:

$$p(y | x) = \frac{p(x | y)p(y)}{p(x)} = \frac{p(x | y)p(y)}{\sum_{y'} p(x | y')p(y')}$$

Law of Total Probability:

$$p(x) = \sum_y p(x, y) = \sum_y p(x | y)p(y)$$

3. D-Separation Rules in Belief Networks

Three canonical 3-node graph structures govern conditional independence in DAGs:

1. **Chain** ($A \rightarrow B \rightarrow C$): Path blocked when conditioning on middle node B ($A \perp\!\!\!\perp C | B$). Unblocked when B is unobserved.
2. **Fork** ($A \leftarrow B \rightarrow C$): Path blocked when conditioning on common parent B ($A \perp\!\!\!\perp C | B$). Unblocked when B is unobserved.
3. **Collider / Immorality** ($A \rightarrow B \leftarrow C$): Path blocked when B and its descendants are unobserved ($A \perp\!\!\!\perp C$). Path **unblocked** when conditioning on B or any descendant of B ($A \not\perp\!\!\!\perp C | B$).

4. Kullback-Leibler (KL) Divergence

Discrete Definition:

$$KL(P \parallel Q) = \sum_x P(x) \log \frac{P(x)}{Q(x)}$$

Continuous Definition:

$$KL(P \parallel Q) = \int P(x) \log \frac{P(x)}{Q(x)} dx$$

Closed-Form Gaussian KL Divergence: For $q(z) = \mathcal{N}(\mu, \text{diag}(\sigma^2))$ and standard prior $p(z) = \mathcal{N}(0, I)$ in d dimensions:

$$KL(q(z) \parallel p(z)) = -\frac{1}{2} \sum_{j=1}^d (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2)$$

5. Jensen's Inequality & ELBO

For any convex function f (or concave function $\log$):

$$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)] \implies \log \mathbb{E}[X] \geq \mathbb{E}[\log X]$$

Evidence Lower Bound (ELBO):

$$\log p(x) \geq \mathbb{E}_{q(z|x)} \left[\log \frac{p(x, z)}{q(z|x)} \right] = \mathbb{E}_{q(z|x)} [\log p(x|z)] - KL(q(z|x) \parallel p(z)) \equiv \mathcal{L}_{\text{ELBO}}$$

6. HMM Dynamic Programming Recursions

Forward Message $\alpha_t(h_t)$:

$$\alpha_t(h_t) = p(v_t | h_t) \sum_{h_{t-1}} p(h_t | h_{t-1}) \alpha_{t-1}(h_{t-1})$$

Backward Message $\beta_t(h_t)$:

$$\beta_t(h_t) = \sum_{h_{t+1}} p(v_{t+1} | h_{t+1}) p(h_{t+1} | h_t) \beta_{t+1}(h_{t+1})$$

Viterbi Max-Product Recursion $V_t(h_t)$:

$$V_t(h_t) = p(v_t | h_t) \max_{h_{t-1}} [p(h_t | h_{t-1}) V_{t-1}(h_{t-1})]$$